

Section 1: Mutation

explain how a range of mutagens operate, including but not limited to: electromagnetic radiation sources, chemicals and naturally occurring mutagens

What are Mutations and Mutagens?

A *mutation* is a change in the genetic code (DNA structure), which alters the sequence of nucleotides, they can occur spontaneously due to errors in meiosis or mitosis, natural degradation over time, or induced from environmental agents known as mutagens. A *mutagen* is any environmental agent that has the potential to cause a mutation. *Mutagenesis* is the process of mutations occurring, that change from the normal base pairs to altered base pairs. A *carcinogen* can cause cancer (mutations in genes that aid the repair of DNA)

Physical Mutagens (electromagnetic radiation sources)

Main sources of radiation are from the higher end of the spectrum: UV rays, X rays and gamma rays. X rays and gamma rays are dangerous because they have ionising radiation (ability to strip electrons from atoms) which causes damage to DNA and produces free radicals (reactive molecules).

UV radiation is non-ionising but penetrates your skin and has enough energy to hit DNA. Because it is non-ionising it can cause a thymine dimer (when 2 thymine nucleotides next to each other to bond together and cause a bulge in DNA). This is a problem when copying DNA or making an amino acid using polypeptide synthesis, and it can also lead to cancer.

Chemical Mutagens

Chemical interact with DNA by tricking the DNA into thinking they're a part of it, by changing the nucleotides or directly getting in between nucleotides. This causes misreading and changes in polypeptide synthesis. Chemical mutagens are often ingested (like alcohol and tobacco) or environmental (like cleaning products, asbestos, lead).

Chemical mutagens include: *Intercalating agents* (slip between two base pairs and alter DNA form at that point), *base analogue* (has a similar form to nitrogen bases, replication of these molecules is implemented instead of original bases) and *reacting chemicals* (react directly with nitrogen bases and chemically modify them). Metals like mercury can cause mutagenesis if individuals, like miners, are exposed to it for prolonged periods of time.

Naturally occurring mutagens (biological mutagens)

Occur at normal levels in the environment and may cause mutations if individuals have prolonged exposure to it, they are living molecules which can cause mutation and cancer. Living viruses replace our DNA with their own, leading to incorrect polypeptide synthesis, bacteria and fungi release toxins that cause free radicals (unstable atoms) that damage cells by knocking DNA out of place.

Compare the causes, processes, and effects of different types of mutation, including but not limited to: point mutation and chromosomal mutation

Point Mutations

- Involves changes in one or few nucleotides and occurs in a certain point in the DNA usually during replication
- Includes different types: substitution, insertion, deletion.

Substitution

- When one base is changed to a different base in a DNA sequence.
- May have no effect or only affects one amino acid

Frameshift Mutations (Insertions and Deletions)

- Insertion is when one base is added to DNA sequence
- Deletion is when one base is removed from the DNA sequence
- Also called frameshift mutations because they shift the “reading frame” of the DNA sequence.
- Can change all amino acids after the mutation which can alter a protein to make it dysfunctional or non-functional.

Forms of point mutations

Type of point mutation	Effect of point mutation
Silent	There is no change to the amino acid sequence in the polypeptide, because the changed codons code for the same amino acid
Missense	DNA change alters one amino acid in DNA sequence (sickle cell anaemia) Changes level and activity of protein. There is a change to one amino acid in the polypeptide
Nonsense	DNA change which creates premature stop codon, creating an incomplete polypeptide (cystic fibrosis) Protein becomes dysfunctional. There is a premature stop codon, which shortens the polypeptide and results in a non-functional protein
Frameshift	DNA change affects every codon beyond point of mutation which significantly changes the amino acid sequencing. Results in a non-functional protein.

Examples of point mutations

Sickle cell anaemia

Sickle cell disease is a condition that involves two alleles: the normal allele and the mutant, sickle allele. The normal allele's DNA codes for the protein Glu, whereas the mutated sickle DNA has a base of adenine instead of thymine which results in the protein Val. The normal allele is required to produce normal haemoglobin proteins, which result in normal red blood cells. Individuals who are homozygous with the mutant allele have abnormal haemoglobin that leads to sickle shaped red blood cells and poor oxygen transport. However, heterozygous individuals produce both normal and abnormal haemoglobin. These 'carriers' have a unique benefit from their one mutant allele – they are less likely to catch malaria, as this infectious parasite dies inside sickle cells.

Double-muscling cattle

Belgian Blue cattle have a distinctive trait known as 'double-muscling' as they produce higher muscle mass than other breeds, and thus higher quantities of lean meat. This trait arose in cattle from a small deletion within the myostatin gene. This deletion resulted in both a frameshift and a premature stop codon. The protein produced from the mutant myostatin gene is very short and does not function. The normal myostatin protein promotes muscle cells to stop growing and so the mutation causes excessive muscle growth.

Chromosomal Mutations

Numerical abnormalities

- Nondisjunction of chromosomes during meiosis, when homologous chromosomes or sister chromatids fail to separate, results in mutant gametes which can result in offspring containing extra or less chromosomes.

Structural abnormalities within chromosomes

- Chromosomes have sections deleted, duplicated, or translocated from one chromosome to another, or chromosomes being inverted to face the opposite direction

Example of numerical and structural abnormalities: Down Syndrome

Chromosomal mutations can occur in the form of numerical abnormalities, which involve an individual having an extra copy of a chromosome or missing a copy of a chromosome. A normal human karyotype has 46 chromosomes, which includes 23 pairs of autosomes and 2 sex chromosomes. The most common numerical abnormality present in live births is Down syndrome (approximately 15 out of 10 000 births). This condition is also sometimes referred to as trisomy 21, due to the presence of three copies of chromosome 21. Down syndrome causes distinctive physical characteristics, varied levels of intellectual disability and an increased risk of certain health problems, such as heart disease. Another type of Down syndrome, called translocation Down syndrome, can occur when an extra chromosome 21 is translocated onto another chromosome, often 13, 14, 15, or 22.

Structural abnormalities of chromosomes

Type	What is it?	Effect
Deletion	Loss of part or all of chromosome	Affects the growth and development of an organism
Replication	Copy of all or part of chromosome.	Increases gene expression, may be either beneficial or harmful depending on the gene.
Inversion	Reverses direction of parts of chromosome.	Lowered fertility due to production of unbalanced gametes
Translocation	Two chromosomes exchange segments with each other.	Interrupts gene regulation and causes some forms of cancer.

Balanced versus unbalanced abnormalities

Balanced structural abnormalities involve no gain or loss of genes as the genetic material has only been rearranged, such as inversion and translocation. However, unbalanced structural abnormalities involve the gain or loss of genes, such as deletion and duplication.

Crossing over and recombination errors

Crossing over is a process that can occur between homologous chromosomes during meiosis I. Pairs of homologous chromosomes align at the centre of the cell during meiosis I. Crossing over typically involves the two inner chromatids of a pair of homologous chromosomes exchanging corresponding genes between the maternal and paternal copy. Crossing over creates recombinant chromosomes that carry different alleles to the parent chromosomes. Crossing over is one of the ways that meiosis increases variation in offspring. If crossing over does not occur in the typical manner, recombination errors can occur, seen in the following examples:

Two homologous chromosomes meet in the centre of a cell during meiosis I, however they do not align correctly which results in the homologous chromosomes exchanging different amounts of genetic material, causing a duplication or deletion mutation in the chromosomes.

During meiosis I, errors can occur if a chromosome pairs with a non-homologous chromosome. Crossing over between these non-homologous chromosomes results in the recombination of genetic material containing different genes, which cause translocation mutations.

Polyploidy

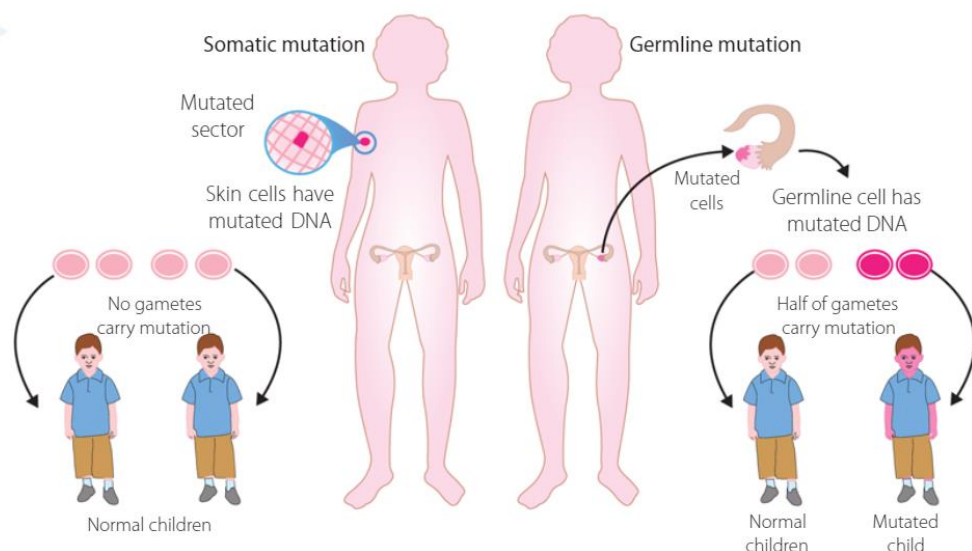
- An inheritable condition of having more than one complete set of chromosomes, which is caused when a mutant diploid gene fuses with normal haploid gene, creating a triploid cell.
- This is very common in plants, such as strawberries, and some polyploid plants are larger, stronger, more resistant to disease and can reproduce normally. Polyploid animals are rare as they usually fail to develop.

Distinguish between somatic mutations and germ-line mutations and their effect on an organism.

Somatic Mutations vs Germline-Mutations

	<i>Somatic mutations</i>	<i>Germline Mutations</i>
<i>Type of cell affected</i>	Body cells (somatic cells)	Sex cells (germline cells produce gametes)
<i>Effect on gene pool</i>	Aren't passed to offspring, hence they don't affect the gene pool	Can be passed to offspring, hence increasing chance for evolution in populations
<i>Causes</i>	Environmental factors	DNA replication errors

FIGURE 7.12
Germline and somatic mutations



Assess the significance of 'coding' and 'non-coding' DNA segments in the process of mutation.

Significance of Coding DNA in Process of Mutation

Coding DNA is DNA sequences that are used directly to make proteins and RNA. Mutations in coding genes affect the type or sequencing of amino acids in a protein. In eukaryotes it can affect gene splicing and modify function or levels of the protein.

Significance of Non-Coding DNA in Process of Mutation

DNA sequences in a non-coding region of DNA control gene expression (where and when genes are expressed), location of splicing to remove introns, and location for ribosome to bind

to mRNA. If mutation occurs in a DNA segment located in this region – production of small nuclear RNA responsible for DNA splicing may be affected. Small nuclear RNA might therefore lose function and hence some introns might not be spliced before translation. Intron which is not spliced away could signal for a stop codon during translation that affects sequencing of polypeptide chain.

Investigate the causes of genetic variation relating to the processes of fertilisation, meiosis, and mutation.

Fertilisation

- Increases genetic variation by combining maternal and paternal genetic material in a zygote, due to independent assortment and random segregation in meiosis, and results in new gene combinations in individuals and the population.

Meiosis

- Meiosis has a significant role in creating genetic variation by introducing new alleles through crossing over (exchanging of genes between homologous chromosomes) and random segregation (two chromosomes separated randomly to reproductive cells' gametes). Replication errors in meiosis can also lead to point mutations, and errors in crossing over or nondisjunction can cause chromosomal mutations.

Mutation

- Mutations can increase the number of alleles for a trait and can occur during the replication of DNA (leading to replication errors) or during the separation (disjunction) of chromosomes (leading to chromosomal mutations).

Evaluate the effect of mutation, gene flow and genetic drift on the gene pool of populations.

A *gene pool* refers to the combination of all the genes (including alleles) present in a reproducing population. A large gene pool has extensive genetic diversity and can better withstand environmental changes.

Gene flow is a term used to refer to changes in allele frequency due to new individuals entering a population or from individuals exiting a population. These individuals can alter the gene pool of the population

Natural selection is a process by which individuals possessing traits that are advantageous for survival in their environment survive in order to reproduce

Sexual selection is a process whereby some traits become more common in a population due to mating partners being selected based on having such traits.

Effect of Mutation on Gene Pool

If the rate of mutation of DNA in germ-line cells increases the frequency of new alleles being passed onto offspring will increase which results in an increase in the gene pool, and visa versa for a decrease in mutations.

Effect of Gene Flow on Gene Pool

Gene flow is the transfer of genetic material from one population to another, through immigration. As species from two populations mate with each other and exchange genetic material, the exchange of alleles increases allele frequency in population, hence increasing genetic variation which results in an increase in gene flow. Alleles may be lost from the population through emigration. This effect is also more noticeable in smaller populations.

Effect of Genetic Drift on Gene Pool

Genetic drift is the change in allele frequency due to chance, which causes individuals to be different, not necessarily more successful. Since one or more alleles are lost in a population due to random events, the allele frequency of gene pool will decrease. The remaining individuals may not be an accurate representation of the genotype frequencies of the original population. The change in allele frequency due to genetic drift is greater if the population is small.

Bottleneck Effect

A type of genetic drift that occurs when a chance event causes a drastic decrease in population size, resulting in genetic drift. The decrease in allele frequency lowers genetic variation, and thus decreases gene pool.

Founder Effect

A type of genetic drift that occurs when a new population is started by a small number of individuals called 'founders' who are not representative of the original population. The new population has a different allele frequency compared to the old population, as founders may not carry all genetic variation of old population, which overall lowers genetic variation in gene pool compared to the old population.

Using population data in conservation genetics

Conservation genetics is the use of data on the DNA and genes of a population to help guide management decisions that seek to preserve the population. Conservation genetic projects usually focus on endangered species and the development of suitable preservation plans.

Genetic diversity assists in the long-term survival of a population. When DNA samples from the population indicate low genetic biodiversity, conservation programs can sometimes include the introduction of new genetic material to the population.

Section 2: Biotechnology

Investigate the uses and applications of biotechnology (past, present, and future), including: analysing the social implications and ethical uses of biotechnology, including plant and animal examples (researching future directions of the use of biotechnology), evaluating the potential benefits for society of research using genetic technologies, evaluating the changes to the Earth's biodiversity due to genetic techniques

Uses and Applications of Biotechnology (Past, Present, Future)

Past Biotechnology Uses/Applications

- **Fermentation** occurs when microorganisms break down sugars during aerobic respiration (commonly used to produce bread, cheese, and yoghurt)
- **Medicines** also used biological processes and organisms (e.g. ancient Egyptians used honey to treat wounds and infections)
- **Selective breeding** humans choose plants or animal individuals to breed together in order to yield offspring with desirable traits (e.g. cattle could be high beef yield)

Modern Biotechnology Uses/Applications

- **Polymerase chain reactions (PCR)** is a genetic technology used to amplify large amounts of DNA, making it available for other applications such as gel electrophoresis.
- **Gel electrophoresis** is used to analyse genetic relationships in a range of applications, including family relationships and forensic investigations.
- **Gene cloning** occurs at a cellular level and involves identical copies of one or more genes.
- **Whole organism cloning** occurs when an entire new identical organism is created.
- **Artificial insemination** involves taking sperm from male and artificially inserting it into females.

Future Biotechnology Uses/Applications

- More **widespread use of cloning and recombinant DNA technology** such as gene therapy (inserting or replacing genes to prevent or treat disease).

Case Study - Penicillin

One of the most famous natural products discovered was an antibacterial product made by the fungus *Penicillium*. The product was discovered in 1928 by English bacteriologist Alexander Fleming. It was later stabilised by Australian pharmacologist and pathologist, Howard Florey, to form the antibiotic known today as penicillin, which is still used for the treatment of bacterial infections. This application revolutionised the treatment of infectious diseases caused by bacteria.

Social Implications and Ethical Uses

Positive Social and Ethical Uses

- **Genetic diversity:** potential to create new arrangements of genes to increase diversity of traits. Has a positive impact upon course of evolution as traits continue to be recombined to best survive in the environment
- **Open-source directions:** Keeping scientific information within the public sphere by creating open-access databases to create a more democratic process such that scientists can work on projects they find to be important.

Concerns with Social and Ethical Uses

- **Ownership:** personal and private information is valuable to large companies who will pay large amounts of money for this information. Gene sequencing can allow people to understand more about themselves as humans however there is a potential for this information to be exploited (e.g. companies who use genetic information to discriminate).
- **Intellectual property:** Allows for certain processes and products to be patented so that people who invented them can make profit. With biotechnology this becomes more complicated, for example questioning the rights a person has to a gene or claiming ownership of a DNA sequencing in a person's body.

Future Directions

- **Synthetic biology:** combines engineering principles with biotechnology.
- Movement started out aiming to characterise the fundamental building blocks of biology, DNA, genes, proteins.
- Progress in this field has been helped by establishment of international research competitions such as **IGEM** (international genetically engineered machines). Capitalises vast creative energy of researchers at the university level and directs them toward developing technologies to address world issues.

Benefits of Biotechnology

- **Pharmaceuticals** such as **vaccines** which use understanding of how the immune system responds to invasion of foreign materials such as viruses, or antibiotics which target certain proteins in bacteria.

- *Stem cell treatments* use therapeutic cloning to harness stem cells to create skin grafts and treat certain cancers or autoimmune diseases.
- *Transgenesis in agriculture* is used to produce genetically modified crops improving the plants ability to survive and increasing nutritional value.
- *Biodegradable plastics* helped to address pollution and improve living systems due to increased biocompatibility.

Effect on Biodiversity

- *Creation of monocultures* use favoured genes that wipe competitive alleles from species gene pools.
- *Horizontal gene transfer* uses the acquisition of genetic information by transfer of a member of different species. As traits escape into ecosystems, this may pose competition to other naturally occurring alleles.

Section 3: Genetic Technologies

Investigate the uses and advantages of current genetic technologies that induce genetic change.

Reproductive Technology

Method	Uses	Advantages
Artificial Insemination	Animal production and fertility treatment	Efficient, makes remote mating possible, can bypass fertility issues.
In vitro Fertilisation	Fertility treatment	Can help patients who otherwise would not be able to conceive.
Artificial Pollination	Pollination of crops	Controlled inheritance of genetic traits.

Cloning Techniques

Method	Uses	Advantages
Whole-organism Cloning	Production of genetically identical offspring.	Definite inheritance of desirable traits.
Therapeutic Cloning	Medicine – stem cell technologies	Stem cells can differentiate into any cell.
Gene Cloning	Medicine and industry	Production of biologically relevant proteins.

Recombinant DNA Techniques

Method	Uses	Advantages
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Transgenesis	Agriculture (development of pest resistant crops etc.)	Creation of organisms with multiple functions, transference of favourable traits, reduction of pesticide use.
Gene Sequencing	Medicine, genetic research.	Identification of genetic disorders and risk factors.
Gene Therapy	Medicine	Treatment of diseases e.g. cystic fibrosis.
ELISA	Medicine	Can identify infections.
CRISPR	Molecular Biology	Treatment of infections and can modify DNA with exceptional precision.

compare the processes and outcomes of reproductive technologies, including but not limited to: artificial insemination and artificial pollination

<i>Technology</i>	<i>Process</i>	<i>Positive Outcomes</i>	<i>Negative Outcomes</i>
Artificial Insemination	1. Extraction of sperm. 2. Insertion into uterus of females.	Favourable genes passed onto offspring	Limits genetic variation
In-vitro Fertilisation (IVF)	1. Hormone treatment which stimulates egg production. 2. Removal of multiple eggs from ovaries 3. Fertilisation (egg and sperm combination) 4. Incubation (leading to production of embryo) 5. If successful, embryo is implanted into the uterus or frozen.	Favourable genes passed onto offspring	Expensive
Artificial Pollination	1. Pollen removed from stamen of one plant. 2. Applied to stigma of another plant. 3. Pollen fertilises ovum.	Selection of desirable traits Creation of new plant species	Loss of biodiversity

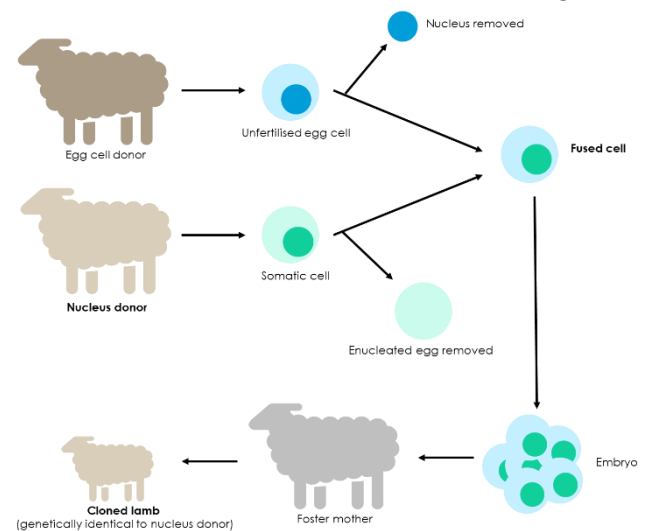
Investigate and assess the effectiveness of cloning, including but not limited to: whole organism cloning and gene cloning.

Whole Organism Cloning

- The creation of new organisms which are genetically identical to their parent.
- Occurs through somatic cell nuclear transfer (creation of a viable embryo from somatic cells and egg cells)

Process

1. Somatic cell is removed from an organism in interest of cloning, as it contains genetic material which is wanted in the offspring.
2. Unfertilised egg is removed from a donor organism; DNA is removed from this.
3. Egg and somatic cell are fused.
4. The resulting cell is cultured so that it begins to divide and become an embryo.
5. Embryo implanted into a surrogate organism.
6. Surrogate gives birth to an organism which is genetically identical to donor.

*Effectiveness*

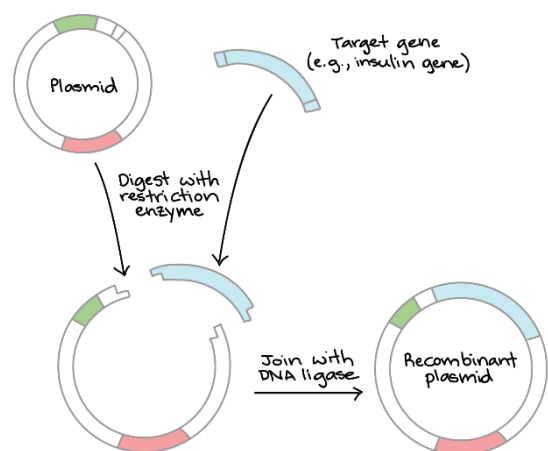
- Clone produced is not strictly identical to parent, if there is a mutation in the somatic cell this is passed onto the offspring.
- Mitochondria present in cytoplasm of donor egg contains DNA which will be passed on to the clone, meaning the cloned organism will have a different mitochondrial genome from its parent.
- Expensive and time consuming.

Gene Cloning

- Production of multiple copies of a specific DNA sequence to produce multiple copies of identical DNA molecules. In other words, to assemble recombinant DNA containing a gene of interest, and then directing this DNA into a host organism which will replicate the gene and produce the target protein in large amounts.

Process (for duplication of DNA by bacteria)

1. A target gene is identified.
2. Plasmids (circular DNA) are isolated from bacteria.
3. Bacterial plasmid and DNA containing the target gene are treated using a restriction enzyme, which cuts DNA and creates 'sticky ends' (sequences of overhanging single stranded DNA).
4. Complementary 'stick ends' of target gene and plasmid come together and recombined using DNA ligase.



5. The new recombinant plasmid is re-inserted into the host bacteria, so that it expresses many copies of the target gene and producing more of the target protein. The protein can be extracted and then used for its purpose.

Effectiveness

- Fast and cheap.
- Harder to produce large amounts of protein on an industrial scale.

Describe techniques and applications used in recombinant DNA technology, for example: the development of transgenic organisms in agricultural and medical applications

Recombinant DNA (rDNA)

Recombinant DNA is defined as the DNA that results from moving a section of DNA from its normal location and inserting it into DNA at another site, which can be within the DNA of the same organism or a different organism. Recombinant DNA can occur naturally, or it can be produced by using biotechnology.

Recombinant DNA for cloning genes in bacteria

Recombinant DNA technology can be used to clone genes in bacteria, this method of gene cloning involves first creating a transgenic bacterium with the desired gene inserted into its DNA plasmid, along with the appropriate gene regulators. When this bacterium undergoes binary fission to reproduce, more transgenic bacteria with the desired trait are produced. Hence each offspring bacterium has a clone of the gene and so can produce the gene product.

This method of gene cloning is used for producing human insulin. The human gene for insulin and a suitable gene regulatory sequence are inserted into an *E. coli* bacterium. This transgenic bacterium reproduces by binary fission, resulting in multiple bacteria, each containing a clone of the gene for producing human insulin. When these transgenic bacteria are grown under ideal conditions, they quickly produce a large amount of human insulin that can be harvested and purified for medicinal use.

Transgenic organisms

Transgenic organisms contain foreign DNA (rDNA), and they are created using genetic engineering to insert one or more foreign genes into an organism's genome. Transgenesis occurs when DNA from a different organism is introduced into a living organism, and as a result the organism exhibits new traits that are transferred to offspring. The insertion of recombinant genes into an organism is possible as the DNA in all organisms has the same molecular structure, featuring the same bases. It is important when inserting recombinant DNA that appropriate gene regulators are included, or the desired gene will not be transcribed.

Methods of Producing Transgenic Organisms

- **Plasmids:** gene editing in bacteria, facilitated by availability of naturally occurring plasmids into which genes may be easily inserted.
- **Retroviral vectors:** retroviruses being exploited to deliver foreign genetic material to a cell. They insert RNA into host cells, which then reverse-transcribe them into their genomes.
- **DNA Microinjection:** Gene of interest is injected into a reproductive cell by a fine glass needle, which is then cultured in vitro until it develops into an embryo. Then this is implanted into the uterus.

Gene Therapy

- When a healthy copy of a gene is inserted into a defective somatic tissue in a developed organism, meaning it cannot be passed onto the next generation.
- Can be transported directly into a cell or via an adeno associated viral vector which causes no diseases and a minor immune response.
- Gene therapy is a new form of medicine, with the potential to replace current treatments for diseases such as genetic disorders.

Elisa (enzyme linked immunosorbent assay)

- An investigative tool that exploits the natural binding of antigens and antibodies. During immune responses, the body produces antibodies specific to invasive antigens.
- By fixing antigens or antibodies to a surface, this can detect whether a corresponding molecule is present in a sample.
- Can be used to test if individuals have produced antibodies specific to an antigen, indicating their exposure to a disease. Can also be used to test presence of antigens in a sample.

CRISPR-Cas9

- The CRISPR-Cas9 system is composed of a guide RNA containing a nucleotide sequence complementary to the gene you want to edit, which is bound to a Cas9 protein which can cut DNA.

- Can be directed to a specific part of a genome with high accuracy to cut and edit the gene of interest. This can substitute, insert, or delete up to 20 base pairs and can improve targeted gene therapy or reversing point mutations.
- Efficient and relatively low cost can improve gene cloning and production of transgenic species.

Evaluate the benefits of using genetic technologies in agricultural, medical and industrial applications

Application	Technique	Benefit
Agriculture	Selective breeding Artificial pollination Transgenesis	Creation of crop and livestock species which exhibit favourable traits Creation of organisms which do not require uses of herbicides or insecticides, thus decreasing impact on environment
Medicine	Gene cloning Gene sequencing Gene therapy CRISPR	Personalised medicine which diagnosis of disorders for better treatment. Improved creation of important biological molecules for treatment of disease.
Industry	Gene cloning Transgenesis	Creation of organisms which produce industrially significant products (biofuels, energy etc.)

Evaluate the effect on biodiversity of using biotechnology in agriculture

Golden Rice (plant biotechnology)

Agricultural applications of biotechnology include the use of genetically engineered crops, such as golden rice, to address hunger in poor and developing countries and work towards eradicating starvation. Golden rice was created by the insertion of two genes, one from the daffodil (replaced by corn in a later version) and one from a bacterium, into a rice genome. Golden rice differs from other rice in that it contains a vitamin A pre-cursor, beta carotene. This rice, with its much higher nutritional (vitamin A) content, was then developed and its creators formed a deal with a company to produce it in large quantities for humanitarian purposes to help combat vitamin A deficiency (which can cause blindness and other serious health problems) in the developing world.

Advantages	Disadvantages
Public health benefits (aid with vitamin A deficiency)	Potential loss of biodiversity in surrounding areas due to unsustainable monoculture practices.
Introduction of new alleles into the gene pool which increases variation across rice species.	Risks of introducing new molecules into a person's diet.
Development of this technology outlines ways for future research into nutritionally augmented food.	Fears of widespread use diverting attention away from continuing structural inequalities causing Vitamin A deficiency.

Sumatran rhino (animal biotechnology)

In animal biotechnology, researchers are mapping relatedness between animals (in the wild and in captivity) and then designing breeding programs to maximise hybrid vigour and genetic biodiversity. Some species that are close to extinction, such as the Sumatran rhino, cannot be saved in the wild. Captive breeding technologies are critical in saving these species. The southern white rhino has been brought back from the brink of extinction by captive-breeding programs. At the start of the 20th century, there were approximately 50 southern white rhinos left. As a result of captive breeding with close mapping of relatedness, there are now over 20 400 white rhinos, with no apparent birth defects despite inbreeding from a small parent population. The rhinos remain in captive populations in conservation centres, to reduce the risk of them being eliminated in one place by disease, natural disaster or poaching.